EX3B: MODEL PLANNING AND BUILDING – REGRESSION

LEARNING OBJECTIVES:

- To understand regression analysis using a real-world Advertising dataset.
- To explore the relationship between advertising expenditure and sales.
- To perform data visualization using pair plots and correlation heatmaps.
- To build Linear, Polynomial, and Support Vector Regression models.
- To evaluate and compare regression models using MAE, MSE, RMSE, and R^2 Score.
- To identify the model with the lowest prediction error.
- To interpret model results for advertising budget decisions.

METHODS / PYTHON LIBRARIES USED:

LIBRARY / METHOD	PURPOSE	REASON TO USE
Linear Regression	Predict Sales	Provides a simple regression model
Polynomial Regression	Model nonlinear relationships	Captures curved relationships
SVR with RBF	Predict Sales	Handles nonlinear patterns
Train-Test Split	Divide data into training and testing	Evaluates model performance
R^2 Score	Measure model fit	Shows how well the model explains Sales

PROGRAM CODE:

```
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import PolynomialFeatures, StandardScaler
```

```
from sklearn.pipeline import Pipeline
from sklearn.svm import SVR
from sklearn.metrics import mean_absolute_error, mean_squared_error,
r2_score

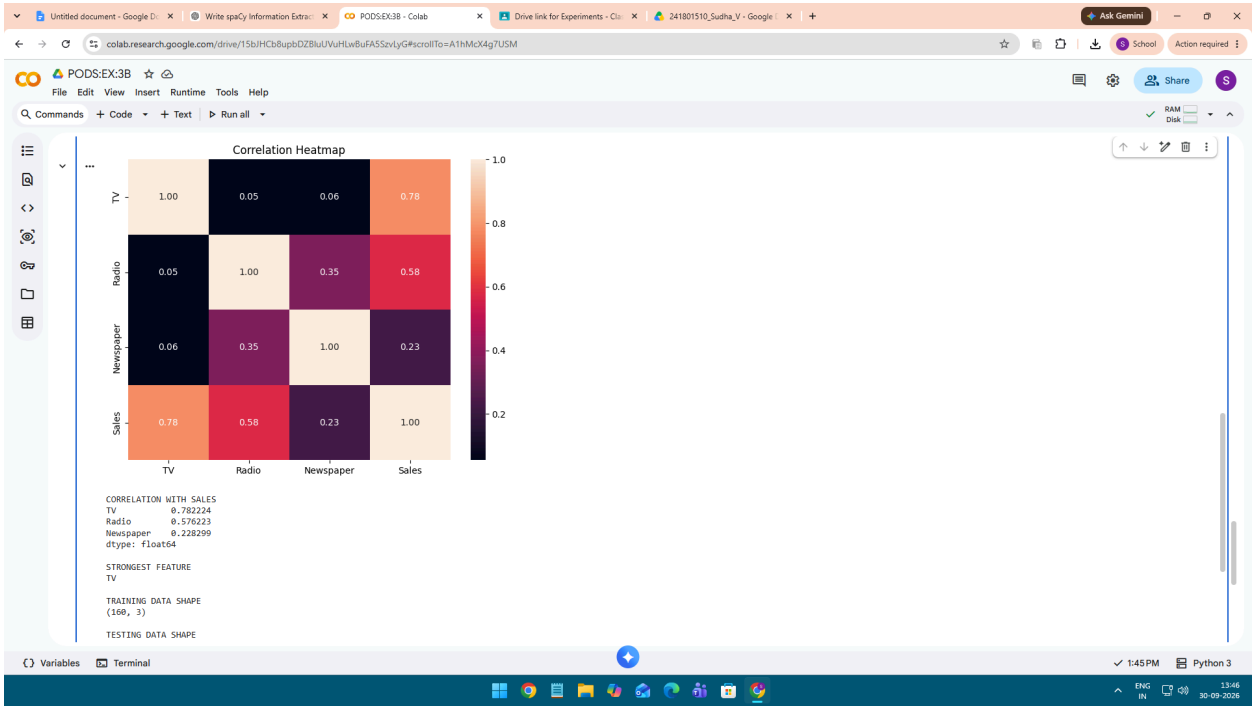
df = pd.read_csv("Advertising.csv")
print(df.head())
print("\nShape:", df.shape)
print("\nMissing Values:\n", df.isnull().sum())
X = df[["TV", "Radio", "Newspaper"]]
y = df["Sales"]
print("\nCorrelation:\n", X.corrwith(y))
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
models = {
    "Linear Regression": LinearRegression(),
    "Polynomial Regression": Pipeline([
        ("poly", PolynomialFeatures(2)),
        ("linear", LinearRegression())
    ]),
    "SVR": Pipeline([
        ("scale", StandardScaler()),
        ("svr", SVR())
    ])
results = []
for name, model in models.items():
    model.fit(X_train, y_train)
    pred = model.predict(X_test)
    mae = mean_absolute_error(y_test, pred)
    mse = mean_squared_error(y_test, pred)
    rmse = np.sqrt(mse)

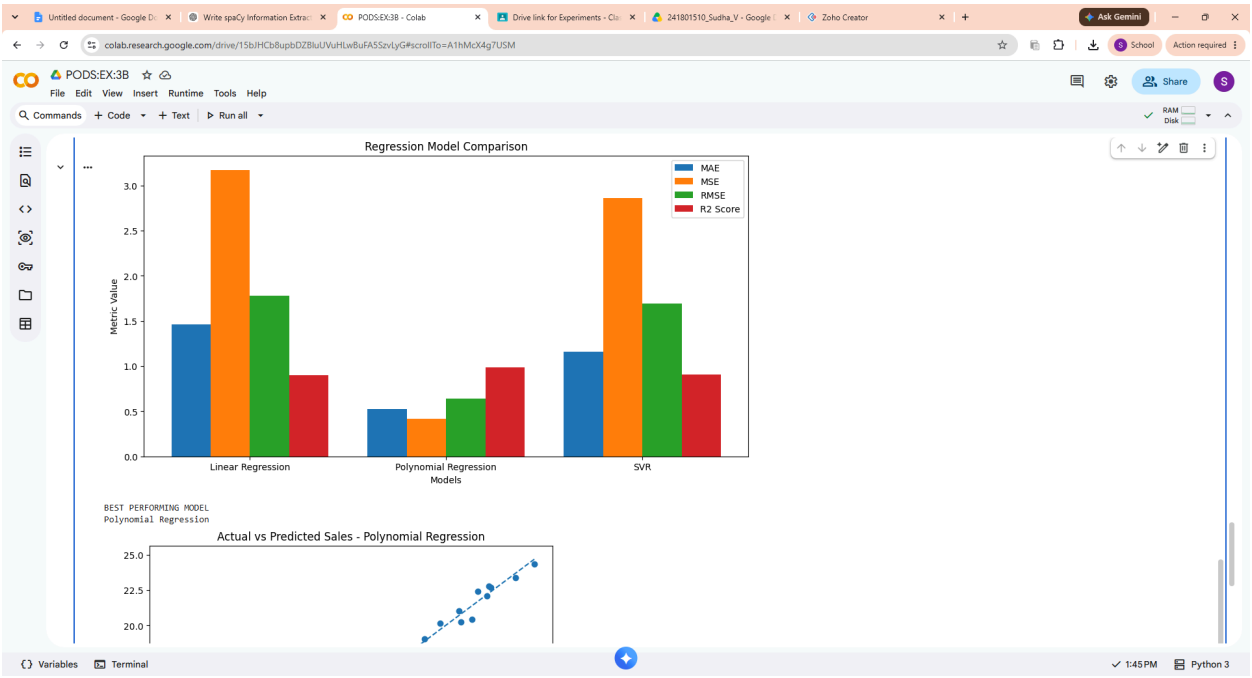
print("\nMODEL COMPARISON")
print(result.round(4))

best = result.loc[result["RMSE"].idxmin(), "Model"]
print("\nBest Model:", best)
```

OUTPUT:


max 200.000000 296.400000 49.600000 114.000000 27.000000
TV
Radio
Newspaper


Correlation Heatmap
TV
Radio
Newspaper
Sales
1.00 0.05 0.06 0.78
0.05 1.00 0.35 0.58
0.06 0.35 1.00 0.23
0.78 0.58 0.23 1.00
CORRELATION WITH SALES
TV 0.782224
Radio 0.576223
Newspaper 0.228299
dtype: float64
STRONGEST FEATURE
TV
TRAINING DATA SHAPE
(160, 3)
TESTING DATA SHAPE



LEARNING OUTCOME:

- Analyze advertising and sales data using Python.
- Identify relationships between advertising channels and Sales.
- Build Linear, Polynomial, and SVR regression models.
- Evaluate regression models using MAE, MSE, RMSE, and R^2 Score.
- Compare different regression models using visualization.
- Identify the model with the lowest prediction error based on the selected evaluation metric.
- Use regression results to inform advertising budget analysis.